

■ B.5 Patterns and Transformation Rules

■ B.5.1 Patterns

Patterns stand for classes of expressions. They contain *pattern objects* which represent sets of possible expressions.

<code>_</code>	any expression
<code>x_</code>	any expression, given the name <i>x</i>
<code>x:pattern</code>	a pattern, given the name <i>x</i>
<code>pattern ? test</code>	a pattern that yields True when <i>test</i> is applied to its value
<code>_h</code>	any expression with head <i>h</i>
<code>x_h</code>	any expression with head <i>h</i> , given the name <i>x</i>
<code>—</code>	any sequence of one or more expressions
<code>___</code>	any sequence of zero or more expressions
<code>x__</code> and <code>x___</code>	sequences of expressions, given the name <i>x</i>
<code>__h</code> and <code>___h</code>	sequences of expressions, each with head <i>h</i>
<code>x_h</code> and <code>x___h</code>	sequences of expressions with head <i>h</i> , given the name <i>x</i>
<code>x_:v</code>	an expression with default value <i>v</i>
<code>x_h:v</code>	an expression with head <i>h</i> and default value <i>v</i>
<code>x_.</code>	an expression with a globally-defined default value
<code>Optional[x_h]</code>	an expression that must have head <i>h</i> , and has a globally-defined default value
<code>pattern..</code>	a pattern repeated one or more times
<code>pattern...</code>	a pattern repeated zero or more times

Pattern objects.

When several pattern objects with the same name occur in a single pattern, all the objects must stand for the same expression. Thus `f[x_, x_]` can stand for `f[2, 2]` but not `f[2, 3]`.

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In a pattern object such as $_h$, the head h can be any expression, but cannot itself be a pattern.

A pattern object such as $x_$ stands for a *sequence* of expressions. So, for example, $f[x_]$ can stand for $f[a, b, c]$, with x being the *sequence* (a, b, c) . If you use x , say in the result of a transformation rule, the sequence will be spliced into the function in which x appears. $g[u, x, u]$ would thus become $g[u, a, b, c, u]$.

When the pattern objects $x_:v$ and $x_.$ appear as arguments of functions, they represent arguments which may be omitted. When the argument corresponding to $x:v$ is omitted, x is taken to have value v . When the argument corresponding to $x_.$ is omitted, x has a *default value* that is associated with the function in which it appears.

<code>Default[f]</code>	default value for $x_.$ when it appears as any argument of the function f
<code>Default[f, n]</code>	default value for $x_.$ when it appears as the n th argument (negative n count from the end)
<code>Default[f, n, tot]</code>	default value for the n^{th} argument when there are a total of tot arguments

Default values.

A pattern like $f[x_, y_, z_]$ can match an expression like $f[a, b, c, d, e]$ with several different choices of x , y and z . The choices with x and y of minimum length are tried first. In general, when there are multiple $_$ or $___$ in a single function, the case that is tried first takes all the $_$ and $___$ to stand for sequences of minimum length, except the last one, which stands for “the rest” of the arguments.

When $x_:v$ or $x_.$ are present, the case that is tried first is the one in which none of them correspond to omitted arguments. Cases in which later arguments are dropped are tried next.

<code>Orderless</code>	$f[x, y]$ and $f[y, x]$ are equivalent
<code>Flat</code>	$f[f[x], y]$ and $f[x, y]$ are equivalent
<code>OneIdentity</code>	$f[x]$ and x are equivalent

Attributes used in matching patterns.

Pattern objects like $x_$ can represent any sequence of arguments in a function

f with attribute `Flat`. The value of x in this case is f applied to the sequence of arguments. If f has the attribute `OneIdentity`, then e is used instead of $f[e]$ when x corresponds to a sequence of just one argument.